Summary statistics for numerical variables

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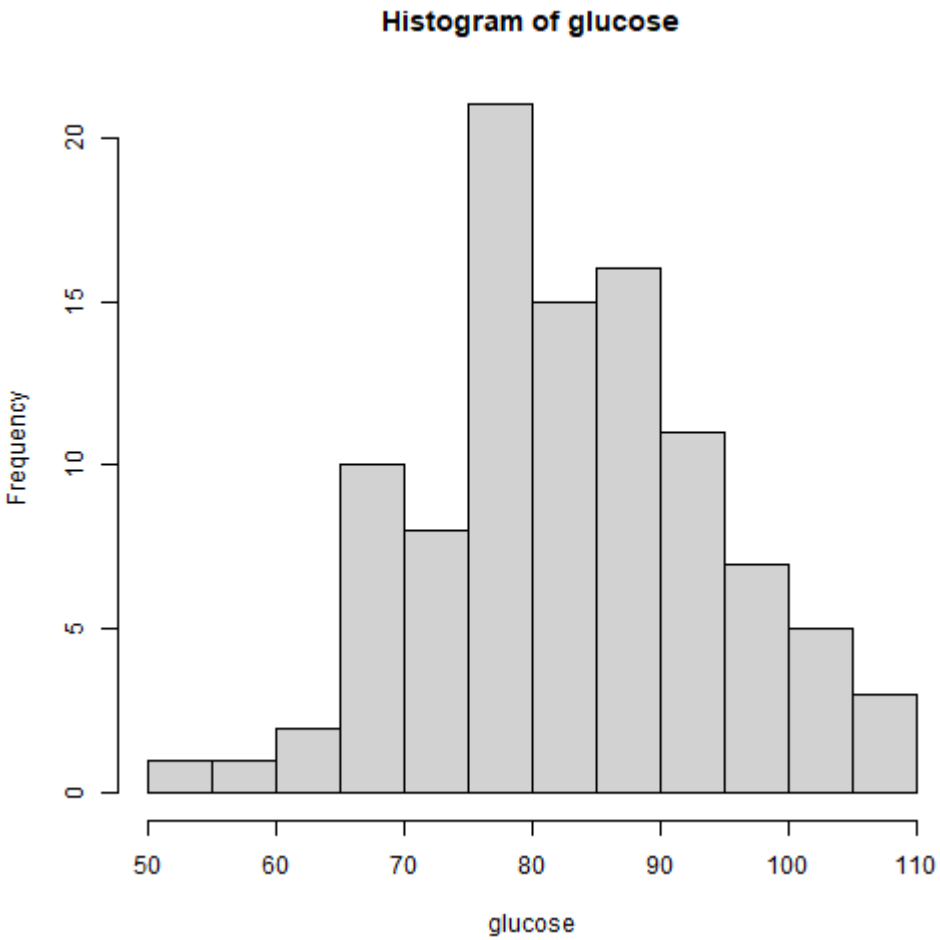
After data collection and cleaning

- The aim is **to summarise the information** in a way that does not influence the real trend below the data
- The representation of the data distribution must describe in **the most unbiased way** as possible
- Indexes of position and dispersion are used, and the extensive description is called exploratory data analysis **(EDA)**
- EDA is one of the most important step **to extract information**

```r
set.seed(123)
glucose <- rnorm(100, mean = 82, sd = 12)
```

```r
hist(glucose)
```



Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

```
mean(glucose)
```

```
## [1] 83.08487
```

Standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

```
sd(glucose)
```

```
## [1] 10.95379
```

Median

Defined for an ordered dataset

If n is odd:

$$\tilde{x} = x_{\left(\frac{n+1}{2}\right)}$$

If n is even:

$$\tilde{x} = \frac{1}{2}\left(x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}\right)$$

```
median(glucose)
```

```
## [1] 82.74108
```

Min Max values

```
min(glucose)
```

```
## [1] 54.28997
```

```
max(glucose)
```

```
## [1] 108.248
```

IQR

- $Q_1 = \text{median of } \{x_{(1)}, \ldots, x_{(\lfloor \frac{n}{2} \rfloor)}\}$
- $Q_3 = \text{median of } \{x_{(\lceil \frac{n}{2} \rceil + 1)}, \ldots, x_{(n)}\}$

$$\text{IQR} = Q_3 - Q_1$$

```
IQR(glucose)
```

```
## [1] 14.22808
```

Summary statistics

The same information can be summarized with these two functions:

```
summary(glucose)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   54.29   76.07   82.74   83.08   90.30  108.25
```

```
summary_stats_gluc <- data.frame(
  Mean = mean(glucose),
  SD = sd(glucose),
  Median = median(glucose),
  IQR = IQR(glucose),
  Min =  min(glucose),
  Max = max(glucose))


summary_stats_gluc
```

```
##       Mean       SD   Median      IQR      Min     Max
## 1 83.08487 10.95379 82.74108 14.22808 54.28997 108.248
```

Creating tables for images and pdfs

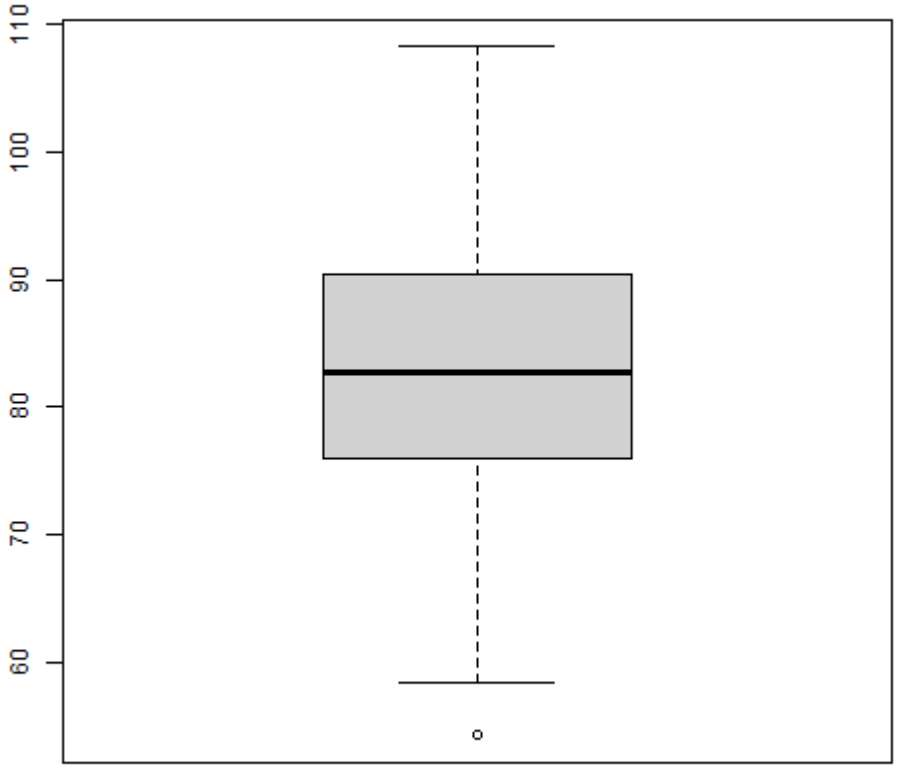
```
library(kableExtra)


summary_stats_gluc %>%
  kbl(caption = "Summary statistics") %>%
  kable_classic(full_width = F, html_font = "Cambria")
```

Summary statistics

Mean	SD	Median	IQR	Min	Max
83.08487	10.95379	82.74108	14.22808	54.28997	108.248

Boxplot

```
boxplot(glucose)
```

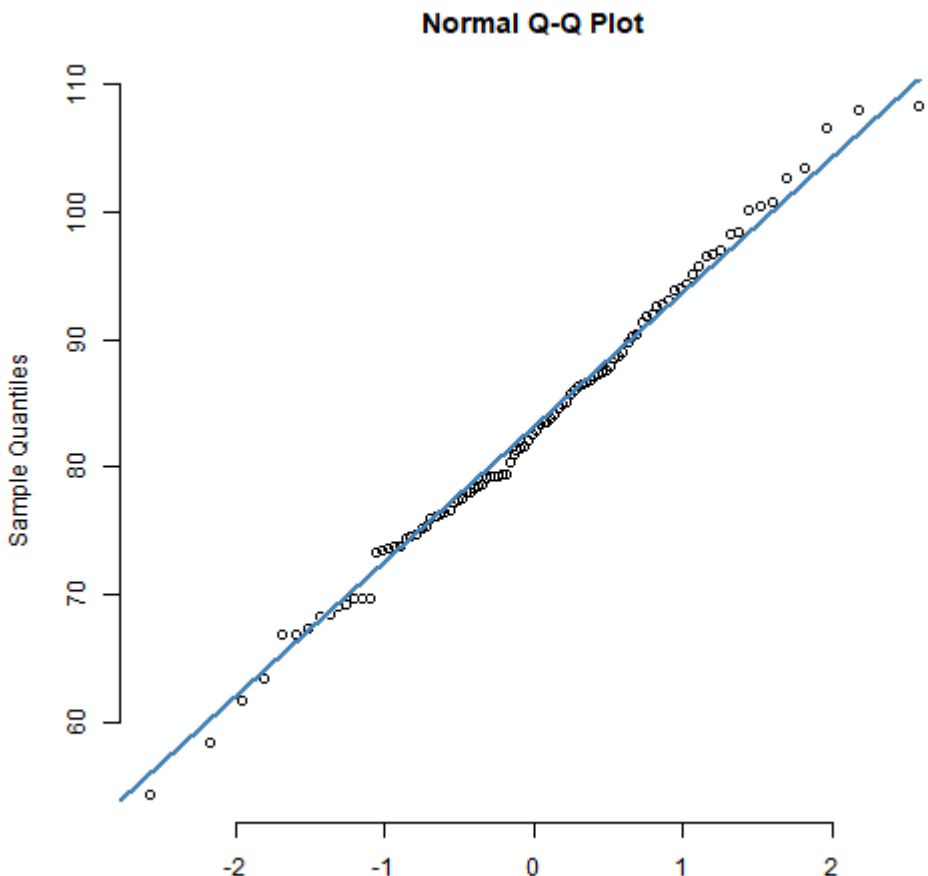


Data distribution

- The distribtuion must be taken into consideration when we have to approach inference.
- We try to understand whether something can be understand about data distrubution for future analysis.
- If data follows the **normal distribution** downstream analysis can be **parametric**
- Otherwise **simmetry** can be checked as well as **kurtosis** and other approaches must be followed (non-parametric).

Visualising normality

```
qqnorm(glucose, pch = 1, frame = FALSE)
qqline(glucose, col = "steelblue", lwd = 2)
```



Testing for normality

```
shapiro.test(glucose)
```

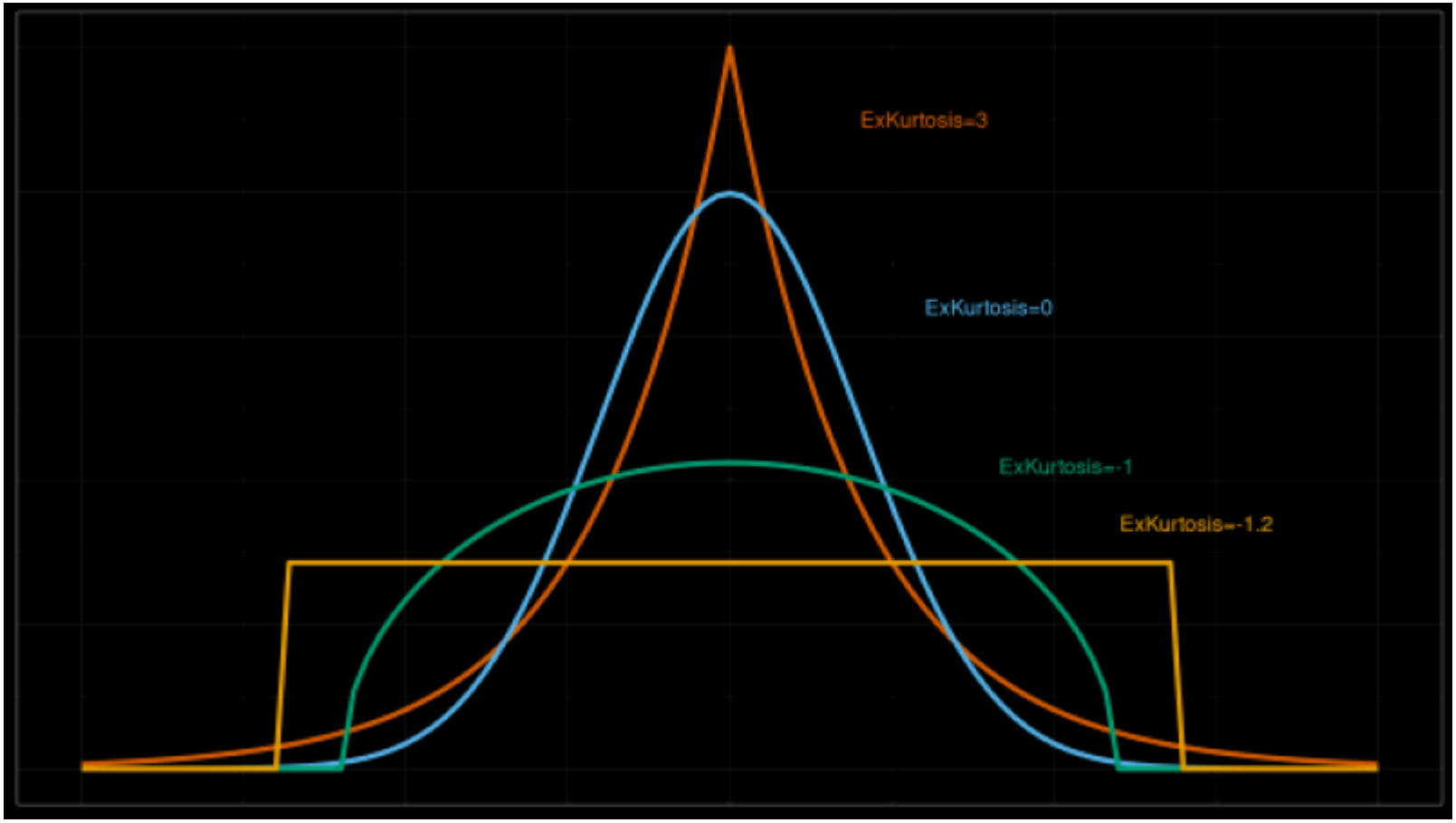
```
##
##     Shapiro-Wilk normality test
##
## data:  glucose
## W = 0.99388, p-value = 0.9349
```

If p-value > 0.05 then parametric tests and methods can be adopted.

Skewness and kurtosis

Skewness: $\frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s} \right)^3$

Kurtosis: $\frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$


ExKurtosis=3
ExKurtosis=0
ExKurtosis=-1
ExKurtosis=-1.2

Estimating skewness and kurtosis

```
library(e1071)
library(moments)
skewness(glucose)
```

```
## [1] 0.06049948
```

- Skewness is the grade of symmetry of distribution distribution.
- Skewness > 0 indicates positive skewness.
- Skewness < 0 indicates negative skewness.
- Skewness = 0 indicates no skewness (symmetrical distribution).

```
kurtosis(glucose)
```

```
## [1] 2.838947
```

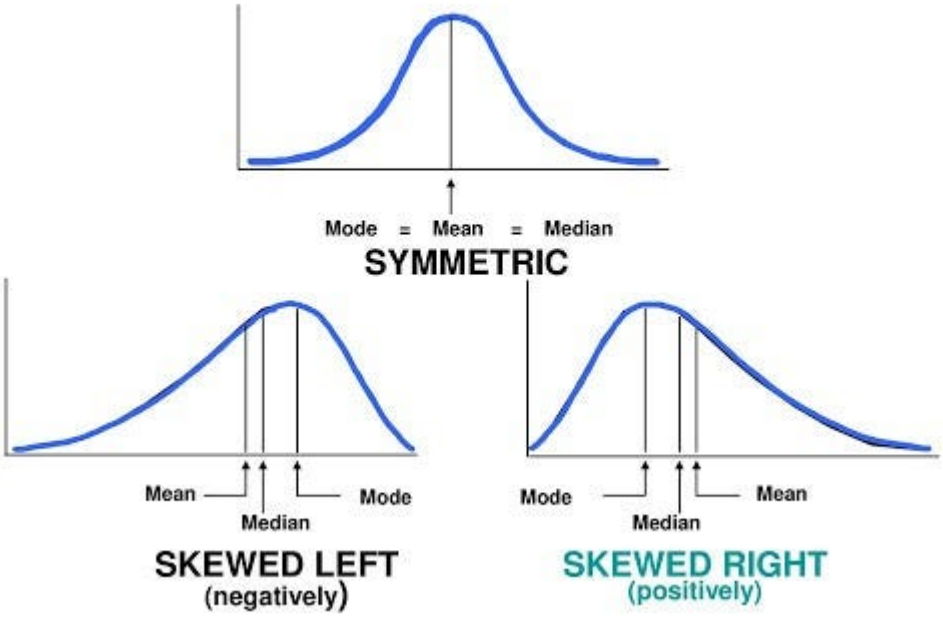
- Kurtosis is the deviation from normal distribution.
- If the distribution deviates from normal distribution we should test whether it is a significant deviation.

Testing for skewness and kurtosis

```r
library(statpsych)

test.skew(glucose)
```

```
##  Skewness      p
##    0.0605 0.7928
```

```r
anscombe.test(glucose)
```

```
##
##     Anscombe-Glynn kurtosis test
##
## data:  glucose
## kurt = 2.838947, z = -0.059837, p-value = 0.9523
## alternative hypothesis: kurtosis is not equal to 3
```

Describe the shape, center, and spread of a distribution… for shape, see below…